

Module/Pathway/Taxon Review:

Tricarboxylic Acid Cycle in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target pathway:** KEGG ppu00020 — Citrate cycle (TCA cycle); module area: central_carbon **Candidate genes reviewed:** 30 (local metadata)

1. Executive Summary

The complete **oxidative tricarboxylic acid (TCA) cycle is encoded and satisfiable** in *Pseudomonas putida* KT2440. All nine steps of the taxon-neutral module — acetyl-CoA entry via citrate synthase, citrate/isocitrate interconversion, isocitrate oxidative decarboxylation, 2-oxoglutarate oxidation, succinyl-CoA-level phosphorylation, succinate oxidation through respiratory complex II, fumarate hydration, malate oxidation, and anaplerotic oxaloacetate input — map to identifiable genes with unambiguous or redundant coverage. There are **no true gaps** in the core cycle.

The single most important lineage-specific feature is the **malate → oxaloacetate node**. KT2440 dedicates three malate:quinone oxidoreductase (MQO) paralogs (*mqa1*/PP_0751, *mqa2*/PP_1251, *mqa3*/PP_2925; K00116, EC 1.1.5.4) to this reaction, while carrying only a single, "putative", non-TCA-bucketed NAD-dependent malate dehydrogenase (*mdh*/PP_0654, K00024). Biochemical evidence from close relatives — *P. aeruginosa* and *P. pseudoalcaligenes* — shows that NAD-MDH activity is often undetectable and that MQO is the physiologically dominant, and sometimes essential, route for malate oxidation. The module's quinone branch should therefore be treated as **primary** in this organism, and PP_0654 flagged `candidate_uncertain`.

The main curation actions are pruning **over-propagated and boundary annotations** that KEGG appends to map00020. Three candidates are methylcitrate/propionate-cycle or CoA-transfer enzymes mis-labelled as core TCA members: *prpC*/PP_2335 is 2-methylcitrate synthase (K01659, EC 2.3.3.5), not citrate synthase; *scpC*/PP_0154 is a succinyl-CoA:acetate CoA-transferase (K18118, EC 2.8.3.18), not the ADP-forming succinyl-CoA synthetase; and both

aconitases (*acnA*/PP_2112, *acnB*/PP_2339) are bifunctional aconitate hydratase / 2-methylisocitrate dehydratase enzymes shared with the methylcitrate cycle. In addition, the pyruvate dehydrogenase complex (*aceE/aceF/acoC*) and the glyoxylate shunt (*aceA*/PP_4116, *glcB*/PP_0356) are neighboring pathways that should be kept as separate modules. Critically, **every KT2440 TCA annotation is homology-inferred** (UniProt protein-existence level 3, or level 4 for pyruvate carboxylase); there is no direct protein- or transcript-level evidence for any individual core gene in this strain.

2. Target-Organism Pathway Definition

2.1 What the module includes

The module covers the **oxidative (clockwise) TCA cycle** as it operates aerobically in *P. putida* KT2440: the eight canonical enzymatic steps from acetyl-CoA + oxaloacetate → citrate around to regeneration of oxaloacetate, plus two curatorial extensions specific to bacterial biology — the **quinone-linked malate oxidation alternative** and **pyruvate carboxylase anaplerosis** feeding the oxaloacetate node.

P. putida KT2440 is an obligate aerobe with a strongly respiratory, oxidative central metabolism. It lacks a canonical Embden–Meyerhof–Parnas glycolysis flux (using the Entner–Doudoroff route and the cyclic EDMP topology instead) and relies heavily on the TCA cycle, glyoxylate shunt, and gluconeogenesis for growth on organic acids. ¹³C flux analysis places the TCA cycle firmly within the tightly-regulated, transcriptionally-invariant core of central carbon metabolism ([PMID: 24951791](#)).

2.2 Neighboring pathways to keep separate

Neighboring process	KEGG map	Genes in candidate list	Curation note
Pyruvate dehydrogenase complex (acetyl-CoA generation)	ppu00785 (lipoic acid)	<i>aceE</i> /PP_0339, <i>aceF</i> /PP_0338, <i>acoC</i> /PP_0553	Feeds acetyl-CoA into the cycle; not a cycle reaction
Methylcitrate / propionate cycle	ppu00640	<i>prpC</i> /PP_2335, and bifunctional <i>acnA/acnB</i>	Over-propagated into map00020
Glyoxylate bypass	ppu00630	<i>aceA</i> /PP_4116, <i>glcB</i> /PP_0356 (NOT in candidate set)	Separate module; anaplerotic on C2 substrates
Acetate/CoA-transfer activation	—	<i>scpC</i> /PP_0154	CoA-transferase, not SCS
Isocitrate DH bucket artifact	ppu04146	<i>icd</i> /PP_4011, <i>idh</i> /PP_4012	Genuinely TCA; KEGG primary-bucket artifact

2.3 Alternate names and database definitions

The pathway is variously called the **citrate cycle**, **citric acid cycle**, **Krebs cycle**, or **TCA cycle**. KEGG defines it as ppu00020 "Citrate cycle (TCA cycle)". BioCyc/MetaCyc split the same chemistry across "TCA cycle I (prokaryotic)" and related variants. The key definitional subtlety for KT2440 is the **isocitrate dehydrogenase cofactor specificity** (NADP-only here) and the **malate-oxidation cofactor** (quinone-linked MQO, not NAD-MDH), both of which distinguish the *Pseudomonas* realization from the *E. coli* textbook version.

3. Expected Step Model and Satisfiability

#	Step	Expected activity	KT2440 gene(s)	KO / EC	Status
1	Acetyl-CoA entry	Citrate synthase (2.3.3.1)	gltA /PP_4194	K01647	covered (single canonical)
2	Citrate → isocitrate	Aconitate hydratase (4.2.1.3)	acnA /PP_2112, acnB /PP_2339	K27802 / K01682	covered (bifunctional caveat)
3	Isocitrate oxidation	Isocitrate DH, NADP (1.1.1.42)	icd /PP_4011, idh /PP_4012	K00031	covered (NADP-only)
4	2-Oxoglutarate oxidation	2-OGDH complex E1/E2/E3	sucA /PP_4189, sucB /PP_4188, E3 (lpd pool)	K00164 / K00658 / K00382	covered
5	Succinyl-CoA → succinate	Succinate-CoA ligase, ADP (6.2.1.5)	sucC /PP_4186, sucD /PP_4185	K01902 / K01903	covered (scpC is a decoy)
6	Succinate oxidation	Succinate DH, quinone (1.3.5.1)	sdhC/D/A/B /PP_4193–4190	K00239–K00242	covered (full operon)
7	Fumarate hydration	Fumarate hydratase (4.2.1.2)	PP_0897 (class I), fumC-I /PP_0944, fumC /PP_1755 (class II)	K01676 / K01679	covered (triply redundant)
8	Malate oxidation	L-malate DH (NAD) or MQO (quinone)	mgo1/2/3 /PP_0751,1251,2925; mdh /PP_0654 (putative)	K00116 / K00024	covered via quinone branch
9	Anaplerotic OAA input	Pyruvate carboxylase (6.4.1.1)	pycB /PP_5346, pycA /PP_5347	K01960 / K01959	covered (two-subunit biotin form)

Bottom line: every module step is satisfiable. The only cofactor/branch decisions a curator must lock in are (i) NADP-only IDH at step 3, and (ii) the quinone (MQO) branch as primary at step 8.

4. Candidate Genes and Evidence

4.1 High-confidence core TCA genes

gltA (PP_4194) — citrate synthase, step 1. KEGG genome-wide link queries return **exactly one** gene for the canonical Si-citrate synthase KO K01647 (EC 2.3.3.1) in organism *ppu*: PP_4194 (Finding F007). This makes *gltA* the unambiguous citrate-synthase call. The second "citrate synthase" in the candidate list, *prpC*/PP_2335, is a decoy (see §5). *gltA* sits in the large *sdhCDAB-sucABlpd-gltA* respiratory operon (PP_4185–4194), consistent with co-regulation of the oxidative cycle.

acnA (PP_2112) and acnB (PP_2339) — aconitase, step 2. Both are present, but both carry bifunctional aconitate hydratase / 2-methylisocitrate dehydratase annotations (EC 4.2.1.3 + 4.2.1.99). *acnB* is explicitly labelled a 2-methylisocitrate dehydratase. Step 2 is covered, but a curator should note these enzymes are **shared with the methylcitrate cycle** and their EC 4.2.1.99 activity is not a TCA reaction.

icd (PP_4011) and idh (PP_4012) — isocitrate dehydrogenase, step 3. Both map to **K00031, NADP-dependent ICD (EC 1.1.1.42)** — *icd* is the dimeric NADP-ICD and *idh* the monomeric NADP-ICD. A KEGG genome check for K00030 (NAD-dependent ICD, EC 1.1.1.41) returns **zero genes** in *ppu* (Finding F007). KT2440 therefore has **NADP-only** isocitrate dehydrogenase, exactly matching the module's "NADP-dependent isocitrate dehydrogenase" step. No NAD-ICD step should be expected.

sucA/sucB + E3 (PP_4189/PP_4188 + lpd pool) — 2-OGDH complex, step 4. E1 (*sucA*, K00164) and E2 (*sucB*, K00658) are single-copy. The E3 dihydrolipoyl dehydrogenase is supplied by **three interchangeable paralogs** all mapping to K00382 (EC 1.8.1.4): *lpdG*/PP_4187, *lpdV*/PP_4404, *lpd*/PP_5366 (Finding F003). This same E3 pool is shared with pyruvate dehydrogenase (*aceE/aceF*) and branched-chain 2-oxoacid dehydrogenase — a normal bacterial arrangement. The E3 paralogs are correctly bucketed to ppu00785, and their presence in the candidate list reflects moonlighting across complexes rather than three separate TCA enzymes.

sucC/sucD (PP_4186/PP_4185) — succinate-CoA ligase, step 5. Canonical ADP-forming SCS α (*sucD*, K01903) and β (*sucC*, K01902) subunits. This is the correct call for step 5; *scpC* is a decoy (§5).

sdhCDAB (PP_4193–PP_4190) — succinate dehydrogenase / complex II, step 6. The complete four-subunit operon: cytochrome b556 (*sdhC*, K00241), membrane anchor (*sdhD*, K00242), flavoprotein (*sdhA*, K00239), and iron-sulfur subunit (*sdhB*, K00240) (Finding F005). Fully covered, quinone-linked (EC 1.3.5.1).

Fumarases (PP_0897, fumC-I/PP_0944, fumC/PP_1755) — step 7. Triply redundant: one class-I fumarate hydratase (PP_0897, K01676) plus two class-II fumarases (K01679) (Finding F005). Class-II ("aerobic/iron-independent") fumarases are the expected aerobic workhorses; the class-I enzyme provides oxidative-stress/iron-limitation backup. Step 7 is robustly covered.

mgo1/2/3 (PP_0751, PP_1251, PP_2925) — malate oxidation, step 8. Three MQO paralogs (K00116, EC 1.1.5.4) constitute the dominant malate-oxidation route (see §4.2). This is the defining lineage-specific feature.

pycA/pycB (PP_5347/PP_5346) — pyruvate carboxylase, step 9. The two-subunit biotin-dependent form: biotin carboxylase *pycA* (K01959) + carboxyltransferase *pycB* (K01960) (Finding F004). Provides anaplerotic oxaloacetate. UniProt protein-existence level 4 ("Predicted") — the weakest evidence tier in the set.

4.2 The MQO malate node (Finding F001)

KEGG ppu00020 assigns **three** genes to K00116 malate:quinone oxidoreductase (*mgo1*, *mgo2*, *mgo3*), whereas only one gene — *mdh*/PP_0654 — carries K00024 NAD-MDH (EC 1.1.1.37), and that gene is annotated "probable/putative" with a primary bucket **outside** the TCA cycle (ppu00566). This gene-dosage asymmetry strongly implies quinone-linked malate oxidation is the physiological default.

Transfer from close relatives is **strong at the genus level**: in *P. aeruginosa*, an *mgo* knockout abolished growth on ethanol or acetate, and "*an NAD-dependent malate dehydrogenase activity could not be detected*" (PMID: [12480887](#)). In *P. pseudoalcaligenes* CECT5344, "*Malate dehydrogenase activity was undetectable, but a malate:quinone oxidoreductase activity coupled to the cyanide-insensitive alternative oxidase was found in membrane fractions*" (PMID: [21178163](#)). Both are same-genus *Pseudomonas* observations, so transfer to KT2440 is reasonable, though direct KT2440 enzymatic confirmation is still lacking (KT2440 itself remains homology-level).

Curation implication: mark the quinone branch of step 8 as the **primary/covered** route; mark *mdh*/PP_0654 as `candidate_uncertain` (present, but likely minor or anabolic rather than the main oxidative-cycle enzyme).

4.3 Evidence tier summary (Finding F005)

All 22 core candidates were checked against UniProt protein-existence levels: **all are level 3 "Inferred from homology"**; *pycA/pycB* are level 4 "Predicted". **None are PE1 (protein-level) or PE2 (transcript-level)**. This means every gene call in this module rests on sequence homology and pathway-database inference, not on direct KT2440 biochemistry.

5. Gaps, Ambiguities, and Likely Over-Annotations

5.1 Over-propagated / decoy annotations to prune (Finding F002)

These candidates are bucketed to ppu00020 but are **not** core oxidative-TCA enzymes:

Gene	Locus	Labelled as	Authoritative call	Action
prpC	PP_2335	"Citrate synthase"	K01659 2-methylcitrate synthase (EC 2.3.3.5)	Reassign to methylcitrate cycle; NOT step 1
scpC	PP_0154	CoA transferase (EC 2.8.3.-)	K18118 succinyl-CoA:acetate CoA-transferase (EC 2.8.3.18)	Acetate activation; NOT step 5 SCS
acnA	PP_2112	Aconitase	K27802 bifunctional aconitase / 2-methylisocitrate dehydratase	Covers step 2 but shared w/ methylcitrate
acnB	PP_2339	Aconitate hydratase B	K01682 bifunctional (EC 4.2.1.3 + 4.2.1.99)	Covers step 2 but shared w/ methylcitrate

Canonical single-copy TCA reactions (*gltA* CS, *sucCD* SCS, *sdhABCD* SDH, *sucAB+lpd* 2-OGDH) contrast sharply with these ambiguous/duplicated entries.

5.2 Boundary pathways (Finding F004)

- **PDH complex** (*aceE*/PP_0339 K00163, *aceF*/PP_0338 K00627, *acoC*/PP_0553 K00627 acetoin-cleaving system) carry primary bucket **ppu00785**, not ppu00020. They generate acetyl-CoA that enters the cycle. *acoC* belongs to a distinct acetoin-cleaving complex. Keep as a separate PDH/lipoic-acid module.
- **Glyoxylate shunt**: KEGG genome checks place isocitrate lyase *aceA*/PP_4116 (K01637) and malate synthase *glcB*/PP_0356 (K01638) in the genome, but **both are absent from the 30-gene candidate list** (Finding F007), confirming the glyoxylate bypass is a genomically and curatorially separate module. This matters because KT2440 routes heavy flux through the glyoxylate shunt and gluconeogenesis when growing on acetate/succinate ([PMID: 41260329](#); [PMID: 41390427](#)).

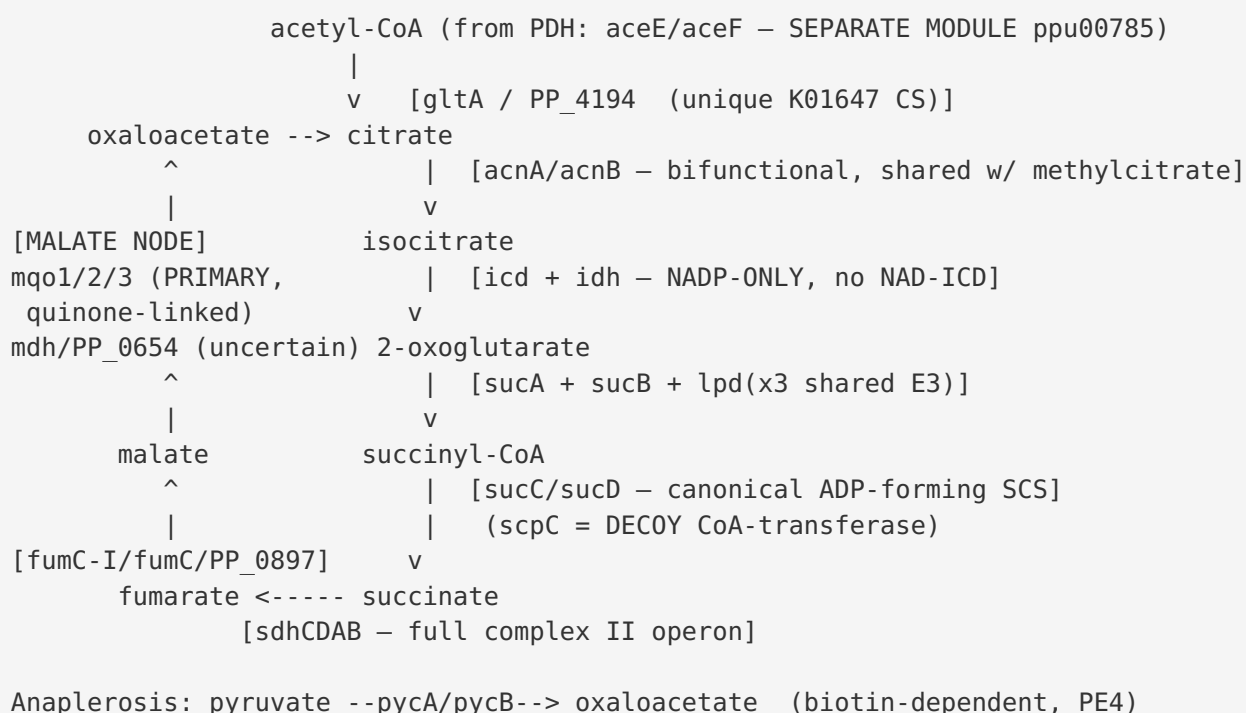
5.3 Genuine ambiguities

- **mdh/PP_0654**: present but "probable", non-TCA bucket, and biochemically often silent in *Pseudomonas*. Status: `candidate_uncertain`.
- **E3 paralog attribution**: three *lpd* genes serve multiple complexes; a curator should not count them as three TCA enzymes.
- **IDH primary bucket**: *icd/idh* are bucketed ppu04146, an artifact; they are genuine TCA step-3 enzymes.

5.4 No true gaps

No expected oxidative-TCA reaction lacks a gene. There is no NAD-ICD (not expected here) and no missing succinate/fumarate/malate node.

6. Mechanistic Model / Interpretation



The KT2440 oxidative TCA cycle is **complete and redundant at three nodes** (malate oxidation: 3 MQO; E3: 3 lpd; fumarate: 3 fumarases) and **single-copy at the diagnostic committed steps** (citrate synthase, SCS, SDH, 2-OGDH E1/E2). The redundancy pattern is itself informative: it concentrates at metabolic branch points that serve multiple pathways (malate node feeds gluconeogenesis; E3 serves three dehydrogenase complexes; fumarases hedge against oxidative stress), while the linear oxidative-decarboxylation core is streamlined. The two cofactor "surprises" relative to the *E. coli* textbook — NADP-only IDH and quinone-linked MQO — reflect *Pseudomonas*'s respiratory, NADPH-provisioning lifestyle and should be encoded as the taxon-default in the module, not treated as anomalies or gaps.

7. Module and GO-Curation Recommendations

Module step	Recommended status	Rationale
1. Citrate synthase	covered	<i>gltA</i> /PP_4194, unique K01647 hit
2. Aconitase	covered (annotate bifunctional caveat)	<i>acnA</i> , <i>acnB</i>
3. Isocitrate DH (NADP)	covered — lock NADP-only	<i>icd</i> , <i>idh</i> ; K00030 absent
4. 2-OGDH complex	covered	<i>sucA</i> , <i>sucB</i> , shared <i>lpd</i> E3
5. Succinate-CoA ligase	covered	<i>sucC</i> , <i>sucD</i> ; exclude <i>scpC</i>
6. Succinate DH (complex II)	covered	<i>sdhCDAB</i> full operon
7. Fumarate hydratase	covered (redundant)	PP_0897 + two class-II fumC
8a. Malate DH (NAD)	candidate_uncertain	<i>mdh</i> /PP_0654 putative, non-TCA bucket
8b. Malate:quinone oxidoreductase	covered — mark primary	<i>mqr1/2/3</i> ; genus biochemical support
9. Pyruvate carboxylase (anaplerotic)	covered (PE4, weak)	<i>pycA</i> , <i>pycB</i>

Module boundary corrections: - Move *prpC*, *scpC* out of the TCA module (into methylcitrate / acetate-activation). - Keep PDH (*aceE/aceF/acoC*) and glyoxylate shunt (*aceA/glcB*) as separate modules. - Set the **quinone-linked malate branch as primary** in the module definition for this taxon; the generic module's default NAD-MDH branch is misleading for *Pseudomonas*.

GO / module-document requests: - No new GO terms are required — existing terms (malate:quinone oxidoreductase activity GO:0008924; isocitrate dehydrogenase (NADP+) GO:0004450; etc.) cover all activities. - Consider a **taxon-note on the module document** flagging (i) MQO dominance and (ii) NADP-only IDH as *Pseudomonas*-characteristic features, so downstream curators do not re-add NAD-MDH/NAD-ICD steps as "gaps".

8. Genes to Promote to Full `fetch-gene` Review

Priority candidates for individual `fetch-gene` review, ranked by curation impact:

1. **mdh / PP_0654** — resolve whether it has any physiological NAD-MDH role in KT2440 or is functionally silent; determines whether step 8a is `candidate_uncertain` or `not_expected`.
 2. **prpC / PP_2335** — confirm 2-methylcitrate synthase identity and formally remove from citrate-synthase step.
 3. **scpC / PP_0154** — confirm CoA-transferase role and remove from SCS step.
 4. **mgo1/2/3 / PP_0751, PP_1251, PP_2925** — determine paralog division of labour (which is the primary TCA MQO vs. condition-specific); relevant to marking a single primary gene.
 5. **acnA / PP_2112 and acnB / PP_2339** — confirm which is the dominant TCA aconitase vs. methylcitrate dehydratase.
 6. **pycA/pycB / PP_5347, PP_5346** — PE4 "Predicted"; verify subunit assignment and anaplerotic role.
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9. Evidence Base (Key Literature)

PMID	Paper (abbrev.)	How it supports / challenges findings
24951791	<i>Functional structure of central carbon metabolism in P. putida</i> KT2440	Direct KT2440: places TCA in tightly-regulated, transcriptionally-invariant central metabolism; supports pathway-level (not per-gene) evidence (F006)
12480887	<i>MQO essential for growth on ethanol/acetate in P. aeruginosa</i>	Genus-level: "NAD-dependent malate dehydrogenase activity could not be detected" — supports MQO-primary malate node (F001)
21178163	<i>Cyanide degradation in P. pseudoalcaligenes CECT5344</i>	Genus-level: "Malate dehydrogenase activity was undetectable, but a malate:quinone oxidoreductase activity...was found" — supports F001
41260329	<i>HexR regulatory landscape in</i> KT2440	Direct KT2440: glyoxylate-shunt/gluconeogenic rerouting on acetate/succinate — supports separate glyoxylate module (F004, F007)
41390427	<i>Ethanol-to-3-HP platform in</i> KT2440	Direct KT2440: genome-scale model shows dominant glyoxylate-shunt flux and oxaloacetate limitation — supports F004
38323821	<i>ML analysis of RB-TnSeq fitness data, P. putida</i>	Resource for resolving per-gene functional confidence (F006) — not yet per-gene mined

Evidence quality note: direct KT2440 evidence exists only at the **pathway/flux/regulation level**. The pivotal MQO-vs-NAD-MDH claim is transferred from same-genus *Pseudomonas* biochemistry (strong genus-level, but not KT2440-direct). All 22 core gene calls are homology-inferred (UniProt PE3/PE4).

10. Limitations and Knowledge Gaps

- **No direct KT2440 protein/transcript evidence.** All core annotations are UniProt PE3 (homology) or PE4 (predicted). Targeted literature searches for KT2440-specific knockout or

biochemical phenotypes of *gltA*, *sdh*, *sucAB*, or *mgo* returned no dedicated studies (Finding F006).

- **Species transfer for the MQO claim** rests on *P. aeruginosa* and *P. pseudoalcaligenes* biochemistry, not KT2440 itself. Genus-level transfer is reasonable but not proven for this strain.
- **RB-TnSeq data are available but not per-gene mined here.** Genome-wide fitness data with ML-derived functional modules exist for KT2440 ([PMID: 38323821](#)); these could confirm essentiality/conditional-essentiality of each TCA gene but were not integrated at the individual-gene level.
- **Paralog division of labour** among the three MQOs, three E3s, and three fumarases is unresolved; the module treats each set as collectively covering its step.

11. Proposed Follow-up Experiments / Actions

1. **Mine existing RB-TnSeq fitness data** ([PMID: 38323821](#)) for per-gene fitness of all 22 core TCA loci across carbon sources — the fastest route to per-gene confidence without new experiments.
 2. **Query enzyme-assay literature / BioCyc** for direct KT2440 MQO vs. NAD-MDH activity measurements; if absent, flag as an experimental gap.
 3. **Targeted deletion / CRISPRi** of *mdh*/PP_0654 vs. *mgo1/2/3* on succinate/malate/acetate to establish the primary malate-oxidation route in KT2440 directly (CRISPRi toolkits for KT2440 are established — [PMID: 39163848](#), [PMID: 36158112](#)).
 4. **Curatorial edits:** implement the §7 status assignments; move *prpC/scpC* out of the module; add the taxon-note on MQO/NADP-IDH.
 5. **Expert question:** confirm whether the three MQO paralogs are functionally redundant or condition-specialized (e.g., stress, alternative electron acceptors).
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*Report generated from a 5-iteration autonomous review. All KT2440 gene-level conclusions are homology-inferred (UniProt PE3/PE4); species-transfer of the MQO claim is genus-level (*P. aeruginosa*, *P. pseudoalcaligenes*). The curation actions in §7–8 are the actionable deliverables.*

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